Sparse Autoencoders Reveal and Control Gender Bias in Vision-Language Model Captioning: A Cross-Lingual Mechanistic Interpretability Study

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Abstract

Vision-language models (VLMs) generate image captions that often reflect societal gender biases, but the internal mechanisms producing this bias remain poorly understood. We apply sparse autoencoders (SAEs) to the internal activations of three VLM architectures—PaLiGemma-3B, Qwen2-VL-7B, and Llama-3.2-11B—to identify and causally intervene on features associated with gender bias in captioning.

Our main findings are threefold. **(1)** Targeted ablation of the top 100 SAE-identified gender features (0.6% of all features) produces a statistically significant change in gender-marked language across all three models, with effect ratios of 1.8–7.1× relative to random feature ablation controls (25 runs each, 500 images, bootstrap CIs excluding zero). **(2)** The *direction* of the intervention effect depends on model architecture: PaLiGemma-3B shows a −16.1% *decrease* in gender terms, while Qwen2-VL-7B (+3.95%) and Llama-3.2-11B (+5.02%) show *increases*—suggesting that smaller captioning models encode gender as *production* features, whereas larger instruction-tuned models encode gender through *regulatory/suppressor* features. **(3)** Cross-lingual analysis across English and Arabic reveals near-zero feature overlap (CLBAS < 0.015), demonstrating that gender bias is encoded through language-specific features, with implications for multilingual debiasing.

To our knowledge, this is the first application of SAEs for mechanistic interpretability of gender bias in VLMs. Code and data are available at `https://github.com/nour-mubarak/mechanistic_intrep`.

1 Introduction

Vision-language models (VLMs) have demonstrated remarkable capabilities in image captioning, visual question answering, and multimodal reasoning liu2023visual,bai2023qwen,team2024paligemma. However, these models inherit and amplify societal biases present in their training data, particularly gender bias in describing people hendricks2018women,hirota2022quantifying. A man on a skateboard is described as "a *man* performing tricks," while a woman in a similar pose is described as "a *person* standing on a board"—the model systematically deploys gendered language for some groups while using neutral language for others.

Prior work has *measured* such biases through caption-level statistics zhao2017men,hendricks2018women and proposed output-level mitigations such as data augmentation or constrained decoding zhao2017men. However, these approaches treat the model as a black box: they quantify *what* the model does but not *why* or *how* it encodes gender information internally.

Mechanistic interpretability—the study of how neural network components implement specific computations—offers a path toward understanding *how* bias is encoded in model representations. Sparse autoencoders (SAEs), recently applied to text-only language models cunningham2023sparse,bricken2023monosemanticity,templeton2024scaling, decompose dense neural

activations into interpretable, monosemantic features. If gender bias is encoded through specific features, these features can be identified, analyzed, and surgically ablated to test causal claims.

In this work, we extend SAE-based mechanistic interpretability to VLMs for the first time, focusing on gender bias in image captioning across four architectures and two languages. Our contributions are:

1. **Methodological:** First application of SAEs for mechanistic interpretability of gender bias in VLMs. We train SAEs on language-model decoder activations from four VLMs (PaLiGemma-3B, Qwen2-VL-7B, LLaVA-1.5-7B, Llama-3.2-11B) and identify gender-associated features via differential activation analysis.

2. **Empirical:** Causal intervention evidence across three architectures showing that targeted ablation of 100 SAE features (0.6% of total) produces statistically significant changes in gender-marked language—with effect ratios of 1.8–7.1× relative to random ablation controls.

3. **Cross-lingual:** First cross-lingual bias alignment analysis (CLBAS metric) across four VLMs, showing near-zero feature overlap between English and Arabic gender representations—implying that debiasing in one language does not transfer to another.

4. **Mechanistic insight:** Discovery that the *direction* of gender-feature influence varies across architectures: smaller captioning models use excitatory gender features (ablation reduces gender terms), while larger instruction-tuned models use inhibitory/regulatory features (ablation increases gender terms).

2 Related Work

Gender bias in VLMs. hendricks2018women documented systematic gender bias in image captioning, showing that models describe men with activity-related terms and women with appearance-related terms. zhao2017men proposed data augmentation to reduce bias in visual semantic role labeling. hirota2022quantifying developed metrics for quantifying bias in VLM-generated captions. These works measure and mitigate bias at the output level but do not explain the internal mechanisms.

Mechanistic interpretability with SAEs. cunningham2023sparse demonstrated that SAEs can decompose GPT-2 activations into interpretable, monosemantic features. bricken2023monosemanticity scaled this to larger models, showing that individual features correspond to semantically meaningful concepts. templeton2024scaling trained SAEs with 34M features on Claude Sonnet, identifying features for safety-relevant concepts. All prior SAE work has focused on text-only models; we extend this to multimodal VLMs.

Causal interpretability. Causal tracing meng2022locating identifies which model components store factual associations by corrupting and restoring activations. Activation patching geiger2021causal tests causal hypotheses about intermediate representations. Our approach is complementary: SAEs decompose activations into interpretable features, and ablation tests whether specific features causally influence output.

3 Models and Data

3.1 Models

We study four VLMs spanning 2.9B–10.6B parameters (Table 1). All models use a vision encoder connected to a language model decoder; we extract activations from the *language model decoder*

layers (not the vision encoder), as these are the layers that generate text tokens.

Table 1: Vision-language models used in this study. d_{model} is the decoder hidden dimension; n_{features} is the SAE dictionary size (8× expansion).

Model	**Params**	d_{model}	**Layers**	n_{features}	**Intervention**
PaLiGemma-3B	2.9B	2,048	18	16,384	L9, L17
Qwen2-VL-7B	7.6B	3,584	28	28,672	L12
LLaVA-1.5-7B	7.1B	4,096	32	32,768	—
Llama-3.2-11B	10.6B	4,096	40	32,768	L20

Layer selection. Intervention layers are selected at approximately the 50th percentile of each model's depth, following evidence that middle transformer layers encode the richest semantic features cunningham2023sparse,templeton2024scaling. For PaLiGemma-3B, L17 (penultimate layer) serves as a negative control.

3.2 Dataset

We use a COCO-derived bilingual image captioning dataset containing 8,092 images with 40,455 English–Arabic caption pairs. Gender labels are derived from caption content using keyword matching (male: 5,562 samples; female: 2,047 samples, reflecting COCO's inherent male bias).

Gender term dictionary. We track 16 binary gender terms: `he, she, him, her, his, hers, man, woman, boy, girl, men, women, boys, girls, male, female`. Following reviewer feedback, we separately track 18 gender-neutral terms (`person, people, individual`, etc.) to detect whether ablation shifts output toward neutral descriptions.

4 Method

Our pipeline has four stages: (1) activation extraction, (2) SAE training, (3) gender feature identification, and (4) causal intervention.

4.1 Activation Extraction

For each model, we extract activations from selected decoder layers using PyTorch forward hooks. Activations are **mean-pooled** across sequence positions to yield one $\mathbb{R}^{d_{\text{model}}}$ vector per image–caption pair.

Table 2: Activation extraction details per model.

Model	**N samples**	**Layers extracted**	**Source**
PaLiGemma-3B	10,000	0,3,6,9,12,15,17	samples.csv
Qwen2-VL-7B	7,609 (EN) / 6,413 (AR)	0,4,8,12,16,20,24,27	gender-labeled
LLaVA-1.5-7B	5,249 (EN) / 4,449 (AR)	0,4,8,...,31	gender-labeled
Llama-3.2-11B	5,249 (EN) / ∼5,000 (AR)	0,5,10,...,39	gender-labeled

4.2 SAE Architecture and Training

We use a vanilla sparse autoencoder with ReLU activation and L1 sparsity:

$$\mathbf{h} = \mathrm{ReLU}(\mathbf{W}_{\text{enc}}\mathbf{x} + \mathbf{b}_{\text{enc}}) \tag{1}$$

$$\hat{\mathbf{x}} = \mathbf{W}_{\text{dec}}\mathbf{h} + \mathbf{b}_{\text{dec}} \tag{2}$$

$$\mathcal{L} = \mathrm{MSE}(\mathbf{x}, \hat{\mathbf{x}}) + \lambda\|\mathbf{h}\|_1 \tag{3}$$

where $\mathbf{W}_{\text{enc}} \in \mathbb{R}^{n_{\text{features}} \times d_{\text{model}}}$, $\mathbf{W}_{\text{dec}} \in \mathbb{R}^{d_{\text{model}} \times n_{\text{features}}}$, and λ controls sparsity (1×10^{-4} for PaLiGemma-3B, 5×10^{-4} for Qwen2-VL-7B).

Training uses AdamW with learning rate 10^{-4}, batch size 256, 50 epochs with early stopping (patience=10), and a 90/10 train/validation split. The expansion factor is $8\times$ throughout.

4.3 Gender Feature Identification

Gender-associated features are identified by **differential activation analysis**:

1. Encode all gender-labeled samples through the trained SAE encoder
2. Compute mean feature activation $\bar{a}_j^{\text{male}}$ and $\bar{a}_j^{\text{female}}$ for each feature j
3. Rank features by $|\bar{a}_j^{\text{male}} - \bar{a}_j^{\text{female}}|$
4. Select top-k features ($k = 100$, i.e., 0.6% of features for PaLiGemma-3B)

4.4 Causal Intervention

The intervention experiment has three phases:

1. **Baseline:** Generate captions for 500 images (seed=42) with no modification
2. **Targeted ablation:** Register a forward hook that zeros the contribution of the $k = 100$ gender features during generation
3. **Random control:** Repeat 25 times with randomly selected features (same k)

Ablation hook. We implement two hook variants:

Full reconstruction (used for PaLiGemma-3B): Replace activations with full SAE reconstruction after zeroing targeted features:

$$\mathbf{x}_{\text{mod}} = \mathbf{W}_{\text{dec}} \cdot (\mathbf{h} \odot \mathbf{m}) + \mathbf{b}_{\text{dec}} \tag{4}$$

where $\mathbf{m}$ is a binary mask with $m_j = 0$ for ablated features.

Residual ablation (used for Qwen2-VL-7B, Llama-3.2-11B): Subtract only the contribution of targeted features from the original activations:

$$\mathbf{x}_{\text{mod}} = \mathbf{x} - \sum_{j \in \mathcal{A}} h_j \cdot \mathbf{W}_{\text{dec}}^{(:,j)} \tag{5}$$

where $\mathcal{A}$ is the set of ablated feature indices.

Justification for two hook variants. The full-reconstruction hook works when the SAE produces high-fidelity per-token reconstructions. For PaLiGemma-3B (cos_sim = 0.9999), this was the case. For Qwen2-VL-7B, the SAE was trained on mean-pooled activations but applied to per-token activations during generation; the distribution mismatch caused the full-reconstruction hook to destroy output coherence (producing 0 gender terms in all conditions). Residual ablation (Eq. 5) avoids this by preserving the original activations and only removing the targeted features' contribution. Importantly, within each model, the same hook variant is used for both targeted and random conditions, so the targeted-vs.-random comparison remains valid.

Table 3: Intervention results across three VLM architectures. All experiments use 500 images, $k = 100$ ablated features, and 25 random control runs. ΔCI is the bootstrap 95% CI on the paired difference (targeted − random) per image. Significance: CI excludes zero for all three models.

Model	Hook	Baseline	Targeted	Change	Random	Spec.	Ratio	ΔCI
PaLiGemma-3B (L9)	Full recon.	1,522	1,277	−16.1%	$-8.7\% \pm 3.9\%$	−7.3 pp	1.8×	
Qwen2-VL-7B (L12)	Residual	1,315	1,367	+3.95%	$-0.56\% \pm 1.14\%$	+4.5 pp	7.1×	
Llama-3.2-11B (L20)	Residual	1,355	1,423	+5.02%	$-0.84\% \pm 0.76\%$	+5.9 pp	6.0×	

Generation. All captions are generated with deterministic greedy decoding (`do_sample=False`, `num_beams=1`, `max_new_tokens=64`) to eliminate sampling variance.

5 Statistical Analysis

Primary metric. Total gender term count across 500 captions, reported as percentage change from baseline.

Length normalization. Gender rate = total gender terms / total tokens, controlling for caption length changes.

Paired statistics. For each image i, we compute $\Delta_i = \text{count}(\text{ablated}_i) - \text{count}(\text{baseline}_i)$. We report the mean Δ with bootstrap 95% CIs (10,000 resamples) and Wilcoxon signed-rank tests.

One-sided Wilcoxon limitation. Our Wilcoxon test uses the one-sided alternative H_1: targeted < baseline (testing for decrease). For models where targeted ablation *increases* gender terms (Qwen2-VL-7B, Llama-3.2-11B), the one-sided p-value approaches 1.0—this does not indicate absence of effect but rather an effect in the opposite direction. For these models, significance is established via bootstrap CIs on the paired difference (targeted − random): if the 95% CI excludes zero, the effect is statistically significant regardless of direction.

Effect specificity. Specificity = |targeted change%|−|random change%|. Ratio = |targeted|/|random|; values > 1 indicate specificity.

6 Results

6.1 Main Intervention Results

Table 3 presents the core findings across all three models with intervention experiments.

All three models show statistically significant targeted effects: the bootstrap 95% CI on the paired difference (targeted − random) excludes zero in every case. Effect specificity ranges from 4.5 to 7.3 percentage points, and effect ratios range from 1.8× to 7.1×.

Direction divergence. The most striking finding is the *direction* of the effect. PaLiGemma-3B shows a −16.1% decrease in gender terms under targeted ablation, while Qwen2-VL-7B (+3.95%) and Llama-3.2-11B (+5.02%) show increases. This suggests two distinct mechanisms of gender encoding:

- PaLiGemma-3B (smaller, captioning-focused): Gender features *produce* gendered output; ablation removes it.

- Qwen2-VL-7B/Llama-3.2-11B (larger, instruction-tuned): Top-differential features *regulate/suppress* gender; ablation disinhibits it.

This is analogous to the distinction between excitatory and inhibitory circuits in neuroscience.

Instruction tuning effect. The two instruction-tuned models (Qwen2-VL-7B, Llama-3.2-11B) show notably tighter random ablation variance (std = 1.14%, 0.76%) compared to PaLiGemma-3B (std = 3.9%), suggesting that instruction tuning creates more stable internal representations where random feature perturbation has less impact.

6.2 Layer Specificity (PaLiGemma)

Table 4: Layer specificity analysis for PaLiGemma-3B. L9 (mid-layer) carries the entire effect; L17 (penultimate) shows no significant change.

Config	Targeted	Random	Wilcoxon p
L9 (mid-layer)	-16.1%	$-8.7\% \pm 3.9\%$	1.65×10^{-21}
L17 (late layer)	-0.8%	$-4.1\% \pm 5.0\%$	0.999 (n.s.)
L9 + L17	-16.1%	$-8.7\% \pm 3.9\%$	1.65×10^{-21}

Gender features are concentrated in the mid-layer (L9): targeted ablation at L17 produces no significant effect ($p = 0.999$), and combining L9+L17 produces identical results to L9 alone. This confirms that gender bias is encoded in specific intermediate layers, not distributed uniformly.

6.3 Per-Term Analysis

Table 5 shows the most affected terms across models.

Table 5: Top per-term changes under targeted ablation. PaLiGemma-3B reduces pronouns; Qwen2-VL-7B/Llama-3.2-11B increase explicit nouns and possessives.

Term	PaLiGemma-3B	Qwen2-VL-7B	Llama-3.2-11B
her	-73.2%	$+4.1\%$	$+11.9\%$
he	-22.6%	$+0.2\%$	-0.2%
his	-0.4%	$+6.3\%$	$+66.3\%$
man	$+10.4\%$	$+53.6\%$	$+4.8\%$
woman	$+9.3\%$	$+43.6\%$	-4.3%
him	-83.3%	$+90.9\%$	$+5.3\%$
men	-33.3%	$+7.5\%$	-43.9%
son	$+2.8\%$	-30.0%	-40.6%

A consistent cross-model pattern emerges: ablation reshuffles *which* gendered terms appear (e.g., pronouns vs. nouns) rather than uniformly increasing or decreasing all terms. For PaLiGemma-3B, pronouns ("her", "him", "she") are dramatically reduced while explicit nouns ("man", "woman") slightly increase—suggesting the ablated features encode pronoun usage patterns rather than pure gender semantics.

6.4 SAE Quality Metrics

SAE quality varies substantially across models. Notably, Llama-3.2-11B achieves significant intervention results (6.0× ratio, CI excludes zero) despite having the lowest-quality SAE (36.6% explained variance, 98.6% dead features). This suggests the intervention is robust to SAE quality and may strengthen with better-trained SAEs.

Table 6: SAE quality metrics. PaLiGemma-3B has the highest quality; Llama-3.2-11B (original) has the lowest. Intervention results are significant across this quality range.

Model	Cos Sim	Expl. Var.	L0	Dead %
PaLiGemma-3B	0.9999	99.8%	7,992	51.2%
Qwen2-VL-7B	0.9965	66.4%	2,049	71.6%
Llama-3.2-11B	0.9956	36.6%	344	98.6%

6.5 Cross-Lingual Analysis

Table 7: Cross-lingual bias alignment. CLBAS ≈ 0 indicates gender is encoded through language-specific features, not shared cross-lingual representations.

Model	CLBAS	EN Probe	AR Probe
PaLiGemma-3B	0.0004	0.810	0.812
Qwen2-VL-7B	0.004	0.918	0.903
LLaVA-1.5-7B	0.015	0.963	0.899
Llama-3.2-11B	0.004	0.994	0.985

All models encode gender in linearly decodable features (probe accuracy > 0.80), but cross-lingual feature overlap is near-zero (CLBAS < 0.015, Jaccard overlap $< 3\%$ at all layers). This means English and Arabic gender biases are encoded through **entirely separate feature sets**—debiasing in English would not automatically reduce bias in Arabic.

7 Discussion

Excitatory vs. inhibitory gender features. The most important finding is the direction divergence across architectures. PaLiGemma-3B (a simpler, captioning-focused model) appears to encode gender through *excitatory* features: these features directly contribute gendered language to the output, so ablation reduces it. Qwen2-VL-7B and Llama-3.2-11B (larger, instruction-tuned) appear to use *inhibitory/regulatory* features: these features modulate or suppress gendered output, so ablation releases it.

This finding has practical implications: a naïve debiasing approach that ablates top-differential features would *reduce* gender bias in PaLiGemma-3B but *increase* it in Qwen2-VL-7B/Llama-3.2-11B. Architecture-specific analysis is essential before deploying SAE-based debiasing interventions.

Hook method confound. We acknowledge that the direction difference could partly reflect the different hook methods (full reconstruction vs. residual ablation). However, both Qwen2-VL-7B and Llama-3.2-11B use the same residual hook and show the same direction, while the full-reconstruction hook *destroyed* Qwen2-VL-7B's output entirely—suggesting the direction difference is primarily architectural. Running PaLiGemma-3B with the residual hook would disambiguate this (see Limitations).

Cross-lingual implications. The near-zero CLBAS scores across all models challenge the assumption that multilingual models share bias representations across languages. Even Qwen2-VL-7B (natively multilingual) shows CLBAS $= 0.004$. This implies that multilingual fairness requires language-specific debiasing interventions.

Mean-pooling trade-off. Our SAEs are trained on mean-pooled activations but applied to per-token activations during generation. For PaLiGemma-3B, this works because the SAE has near-perfect reconstruction on pooled data. For Qwen2-VL-7B/Llama-3.2-11B, the mismatch necessitated residual ablation (eq:residual). Training per-token SAEs would avoid this mismatch but at significantly higher computational cost.

8 Limitations

1. **SAE training scale:** SAEs trained on 5K–10K samples per model; larger training sets may yield better features.
2. **Binary gender:** We track binary gender terms only; non-binary terms are tracked separately but not used as the primary metric.
3. **Hook inconsistency:** PaLiGemma-3B uses full reconstruction; Qwen2-VL-7B/Llama-3.2-11B use residual ablation. Running PaLiGemma-3B with residual ablation would test whether the direction difference is architectural or methodological.
4. **SAE quality variance:** Llama-3.2-11B's SAE has only 36.6% explained variance. The tuned SAE (99.9% EV) exists but was not used for intervention.
5. **No human evaluation:** We measure gender term counts but do not evaluate caption fluency or semantic accuracy after ablation.
6. **Feature polysemanticity:** We cannot claim the ablated features *exclusively* encode gender; they may be polysemantic.

9 Conclusion

We have demonstrated that sparse autoencoders can identify and causally intervene on gender bias features in VLM internal representations. Across three architectures, targeted ablation of just 0.6% of SAE features produces statistically significant changes in gender-marked captioning language, with effect ratios of 1.8–7.1× relative to random controls. The discovery that different architectures use fundamentally different gender encoding mechanisms—excitatory vs. inhibitory—provides new mechanistic insight into how VLMs process social attributes and has practical implications for bias mitigation strategies.

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A Additional Per-Term Analysis

Complete per-term breakdown for all models is available in the supplementary materials and at `https://github.com/nour-mubarak/mechanistic_intrep`.

B SAE Training Details

Table 8: SAE training hyperparameters.

Parameter	**Value**
Expansion factor	8×
L1 coefficient (λ)	10^{-4} (PaLiGemma-3B), 5×10^{-4} (Qwen2-VL-7B)
Learning rate	10^{-4}
Optimizer	AdamW
Batch size	256
Epochs	50 (early stopping, patience=10)
Warmup steps	1,000
Train/Val split	90/10

C Random Ablation Distributions

See supplementary Figure S2 for the distribution of all 25 random ablation runs per model, showing that random effects are centered near zero while targeted effects fall well outside the distribution.